FPT Education

FPT UNIVERSITY

MMA301

Multiplatform Mobile App Development

Semester: FA26	**Curriculum Code**:
Date: 29/10/2026	**Duration**: 85 Minutes

Total Points: 10

The Electric Vehicle Charging Station Management App helps users manage and monitor electric vehicle charging stations. The application allows users to add, update, and delete charging station information, as well as view available charging stations and manage station status. All data must be stored locally using SQLite or AsyncStorage, and the application must run on both Android and iOS via Expo.

Data Structure

Each charging station includes the following attributes:

Field	Type	Description
id	integer	Unique identifier (auto-incremented), e.g., 1, 2, 3
station_name	text	Name of charging station, e.g., "Can Tho Fast Charge 01", "EV Hub Station CT02"
location	text	Address or location of charging station, e.g., "Ninh Kieu District", "Cai Rang District"
charging_power	integer	Charging power in kW, e.g., 60, 120, 150
status	text	Current station status, e.g., "Available", "Occupied", "Maintenance", "Offline"

Business Note:

A charging station with status = "Offline" cannot provide charging service. Therefore, according to the business rule in Task 5, its charging_power must be stored as 0.

General Requirements:

- A **Resources** folder containing sample data is provided. The project must include the required images and provided assets.
- Project folder name: **su26_studentID** (replace *studentID* with your actual student ID).
- Development Tool: **Visual Studio Code**.
- Technology: Use **Expo** and NPM packages to build the application.
- Data Storage: Use **SQLite** or **AsyncStorage**.
- Use **Bottom Tab Navigator** to switch between screens.
- UI Enhancement: Apply proper styling and a responsive layout.

Tasks to be Completed:

Task 1: Charging Station List Screen (Home Screen) (2.0 points)

Create the main screen (see **Figure 1**) displaying all charging stations. This screen must include:

- Display charging stations in a FlatList, showing: Station Name, Location, Charging Power, Status (0.5 point)
- Use Bottom Tab Navigator to switch between screens. (0.5 point)
- On first launch, the list may be empty. Users can add new charging stations using a button. (0.5 point)
- Show a banner image at the top of the Home screen. (0.25 point)
- Tapping a charging station item navigates to its Update screen (0.25 point)

Task 2: Add New Charging Station Screen (2.0 points)

Create a screen to add a new charging station (see **Figure 2**).

- Include fields: *Station Name, Location, Charging Power, Status*

Notes:

- The id field is auto-generated.
- Charging Power must be greater than 0 unless Status is Offline.
- Status must be one of: Available, Occupied, Maintenance, Offline

Requirements:

- Validate all required fields before saving (see **Figure 2**). (0.5 point)
- Save the new charging station to the database. (0.5 point)
- After saving, immediately update the Home screen list. (0.5 point)
- Pressing the Back button should return to the Home screen. (0.5 point)

Task 3: Update Charging Station Screen (2.5 points)

Create a screen for updating (see **Figure 3**) an existing charging station.

Requirements:

- Display the selected charging station's current information in editable input fields.(0.5 point)
- Include an Update button to save modified data and return to the Home screen. (0.5 point)
- Include a Delete button to permanently remove the charging station. (0.5 point)
- Include a Back button to return to the Home screen. (0.5 point)
- Display a confirmation dialog before deleting a charging station. (0.5 point)

Task 4: Available Charging Stations Screen (2.0 points)

Create a screen that displays only charging stations with status = "Available" (see **Figure 5**).

Requirements:

- Display all available charging stations using a FlatList. Show: Station Name, Location, Charging Power, Status. (1.0 point)
- Calculate and display: Total number of available charging stations. (0.5 point)
- Calculate and display: Total charging capacity of all available charging stations. (0.5 point)

Task 5: Additional Requirements (1.5 points)

Implement the following business rule: When status is set to **Offline**, the application must automatically store **charging_power = 0** regardless of the user-entered value. This rule must be applied in both Add and Update operations (*see **Figure 6***).

Requirements:

- Apply the rule when adding a new charging station. (0.5 point)
- Apply the rule when updating an existing charging station. (0.5 point)
- UI/UX: Clean, modern interface with consistent colors, icons (e.g., from @expo/vector-icons), and spacing. (0.5 point)

Examples:

Selected Status	User-Entered Charging Power (kW)	Charging Power Stored in Database (kW)
Offline	150	0
Offline	50	0
Offline	0	0

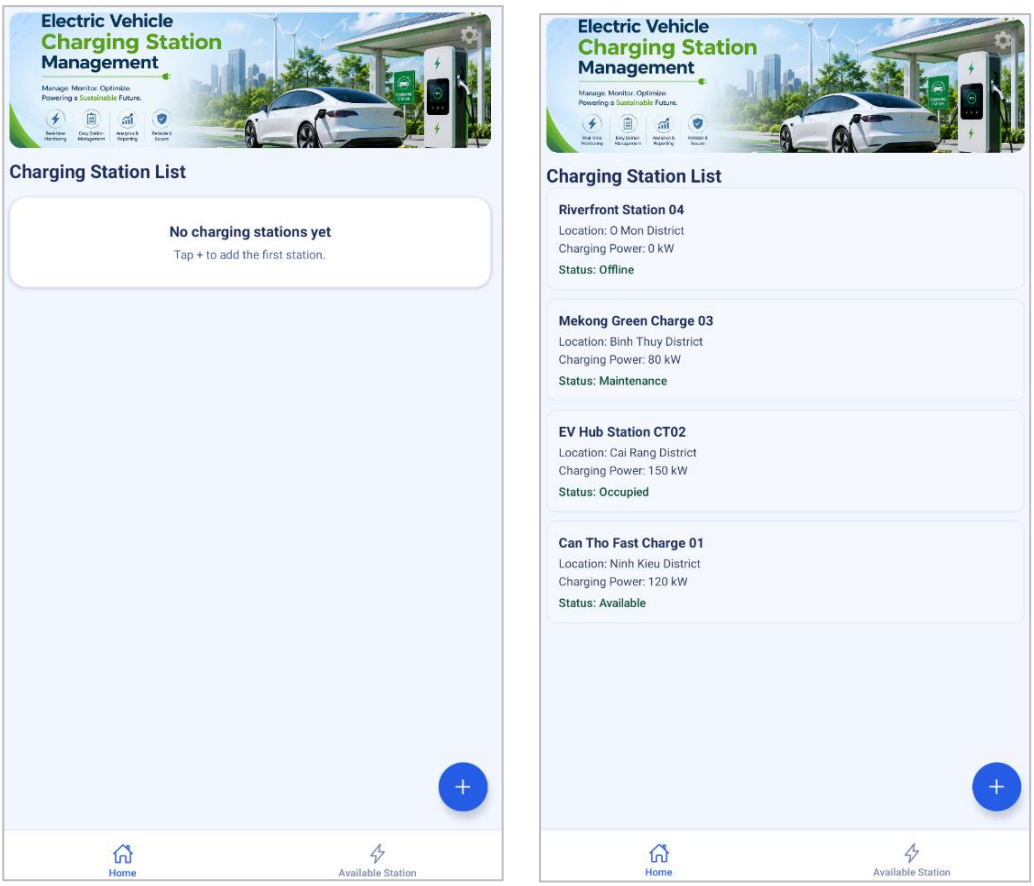


Figure 1: *Home Screen before and after adding charging stations*

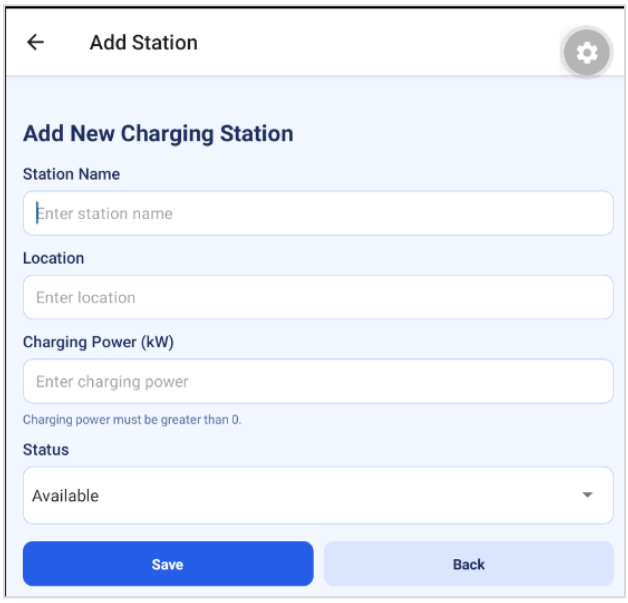


Figure 2: *Add Screen for creating a new charging station, including input fields, validation, and a submit button.*

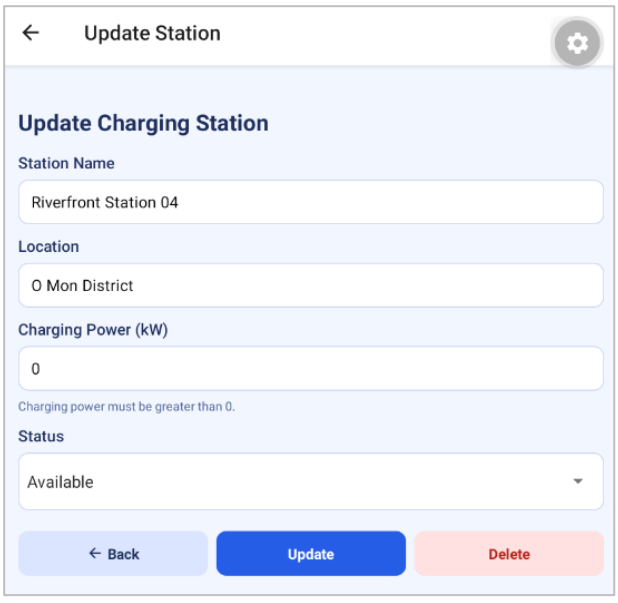


Figure 3: *Update Screen for editing charging station information with options to update or delete the selected charging station.*

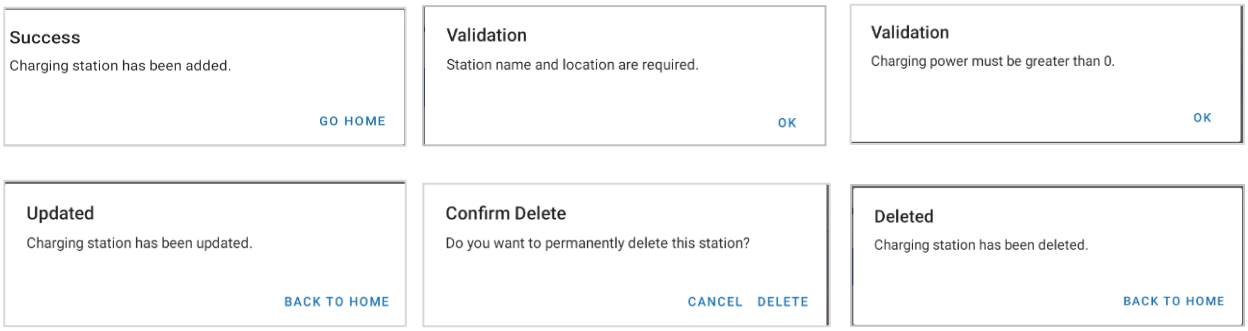


Figure 4: *Screens showing input validation (empty, >0 required fields), delete confirmation dialog, and success messages for add, update, and delete operations.*

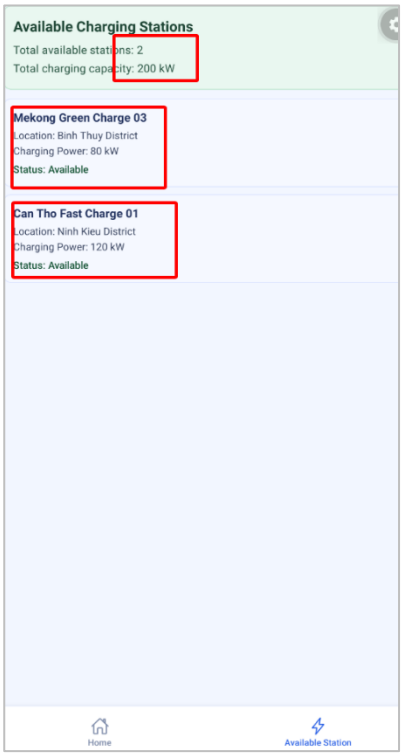


Figure 5: *Available Charging Stations Screen – displays charging stations with status = "Available", along with the total number of available charging stations and the total charging capacity.*

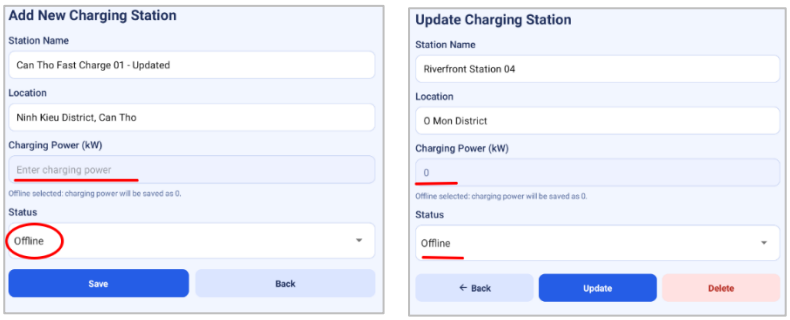


Figure 6: *Station Status Management Rule: when status is set to "Offline", the application must automatically save charging_power as 0.*